



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 · STANDARD 6.GR.4

Surface Area of Prisms

Lesson 10-4 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can find the surface area of rectangular and triangular prisms.

Language Objective: I can explain my work using the words surface area, rectangular prism, triangular prism, and face.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Surface area

Rectangular prism

Triangular prism

Face

Net

Lateral face



Tap any word to see what it means and a picture.

1 The total area of all the faces of a solid is its .

2 A solid with six rectangular faces is a .

3 A solid with two triangular faces and three rectangular faces is a

.

4 A flat surface of a solid figure is a .

5 A flat pattern that folds into a solid is a .

6 A side face of a prism that is not a base is a .

Reflect — Exit Ticket

A rectangular prism has $l = 8$ ft, $w = 6$ ft, $h = 3$ ft. What is the surface area?

- A. 180 ft^2
- B. 144 ft^2
- C. 180 ft^3
- D. 90 ft^2

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Surface area
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Triangular prism
4. **Notes 4:** Face
5. **Notes 5:** Net
6. **Notes 6:** Lateral face
7. **Watch for this mistake:** *A common mistake in Surface Area of Prisms is skipping the key idea: "No matter the prism, find the area of every face and add them all together." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 148 cm^2 — $SA = 2(6 \times 5) + 2(6 \times 4) + 2(5 \times 4) = 60 + 48 + 40 = 148 \text{ cm}^2$.
9. **Try It 2:** A. 288 in^2 — *Two triangle ends: $2 \times (\frac{1}{2} \times 8 \times 6) = 48 \text{ in}^2$. Three rectangles: $(8 + 6 + 10) \times 10 = 24 \times 10 = 240 \text{ in}^2$. $SA = 48 + 240 = 288 \text{ in}^2$.*
10. **Exit Ticket:** A. 180 ft^2 — $SA = 2(8 \times 6) + 2(8 \times 3) + 2(6 \times 3) = 96 + 48 + 36 = 180 \text{ ft}^2$. *Surface area uses square units (ft^2).*